

Functions

A function is a self-contained block of code designed to perform a specific task. It takes inputs, processes them. And Optionally returns an Output.

```
public static void main(String[] args) {
```

```
// Functions in Java (Function hamesha PSVM ke bahar banega.)

//1st call
greet();

//2nd call
sayHello("Priyansh"); // yaha pr (priyansh) ek Argument ha.
//3rd call
int x = getNumber();
System.out.println(getNumber());
//4th call
multiply(12,3);
System.out.println(multiply(12,3));

}

/* Type of Functions---> there are "4" type of functions.
Type-1) No input, No output. */
static void greet() { // jab bhi {no in, no op} hota ha too vha pr {Return type}-->
"VOID" lagate ha.
    System.out.println("Hello");
    // return; // ye bhi optional ha.
}

// Type-2) Input, No Output.
static void sayHello(String name) { // yaha input ha ---> (String name).
    System.out.println("Hello " + name);
}

// Type-3) No Input, Output.
static int getNumber(){// yaha pr output ha---> {return 10;}.
    return 10;
}

// Type-4) Input, Output.
static int multiply(int a, int b){// yaha pr input-->(int a, int b), or output ha-->
{return (a*b)}

    return (a * b);
}
```

Function Overloading....

```
public class FunctionOverloading {
    public static void main(String[] args) {
        // Function Overloading:
        int x = sum (3, 5);
        System.out.println(x); // hum normaly functin ko is tarah se use krte ha .

        // ab me agar 3 digits ko add kru to me niche function add kr skta ha
        int y = sum(3444, 432, 333);
        System.out.println(y);
        int p = sum( 23.3, 34);
        System.out.println(p);
    }
}
```

```

        Out of the main
    }

    static int sum(int a, int b){
        return(a + b);
    }

    // ab agar muj 3 values ko add krna ha to me asa krk skta hu.
    static int sum(int a, int b, int c) { // kuki beshak sum same ha pr Arguments to alag
        ha.
        return (a + b + c);
    }

    // or me different type of parameters bhi rakh sakta hu.
    static int sum(double r, double s){
        return (int)(r + s);
    }

```

Changing of functions...

```

    public class FunctionChanning {
        public static void main(String[] args) {

            // Channing in function---> iska concept same ha Nested ki tarah .
            function1();
            System.out.println("bye");

        }

        static void function1(){
            function2();
            System.out.println("okay");
        }

        static void function2(){
            function3();
            System.out.println("Hello");
        }

        static void function3(){
            System.out.println("Hii");
        }

    }

```

Scope of variables....

```

    // One more important topic--> Scope of variable:
    //it means ki koi bhi variable ha wo agar "{}" ke ander ha to vo sirf wha tk hi
    simit ha.

```

Recursion:

It simply means calling itself.

Example: printnum

```

public static void main(String[] args) {
    // Recursion in function-->

    int n = 98;

```

```

printNum(n);

}

static void printNum(int n) {
    if (n==0) return;
    printNum(n-1); // Recursive call
    System.out.print(n+", ");
}

```

Printing Fibonacci numbers by Recursion —>1, 1,2, 3, 5, 8,

```

public static void main(String[] args) {
    int n = 6;

    System.out.println("Fibonacci sequence upto " + n + ":");

    for(int i = 1; i<= n; i++){
        System.out.print(fibonacci(i) + ", ");
    }

    }

    static int fibonacci (int n){
        if(n<=1){
            return n;
        }
        return (fibonacci(n-1) + fibonacci (n-2));
    }
}

```